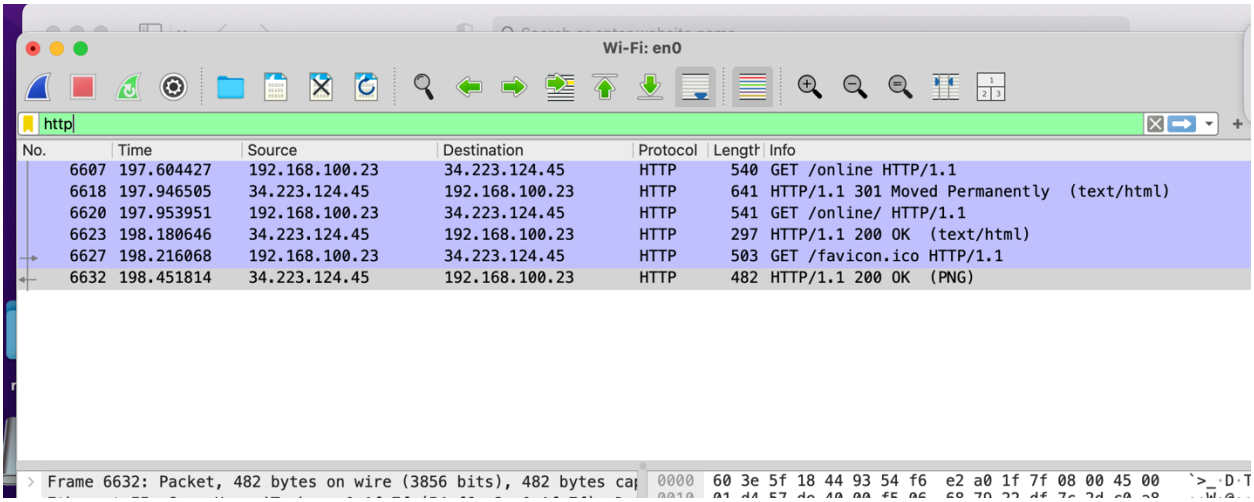
CS471 – Web Technologies (Laboratory)
Lab 1 – The Internet Protocols
Part 4: Comparing TCP and UDP

Name: Remas Turki alquhidan
Student ID: 432205422

Part 1: HTTP Traffic Analysis:

Figure 1: Captured HTTP request and response packets using Wireshark.



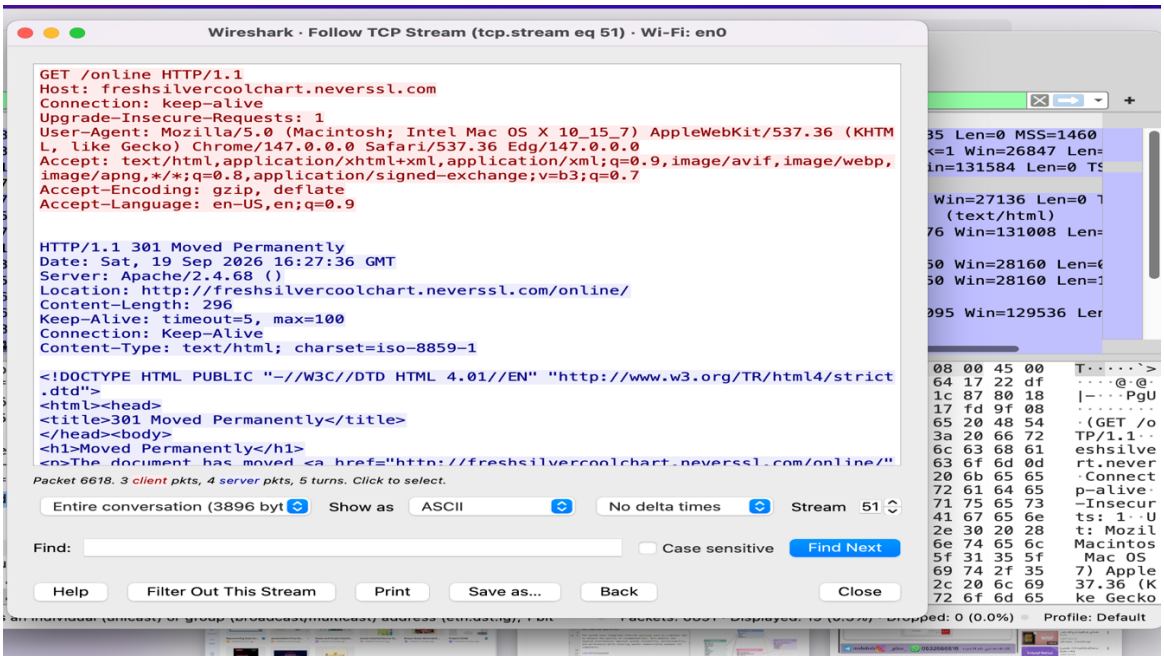


Figure 2: TCP stream showing the HTTP request and server response.

Part 2: TCP Analysis:

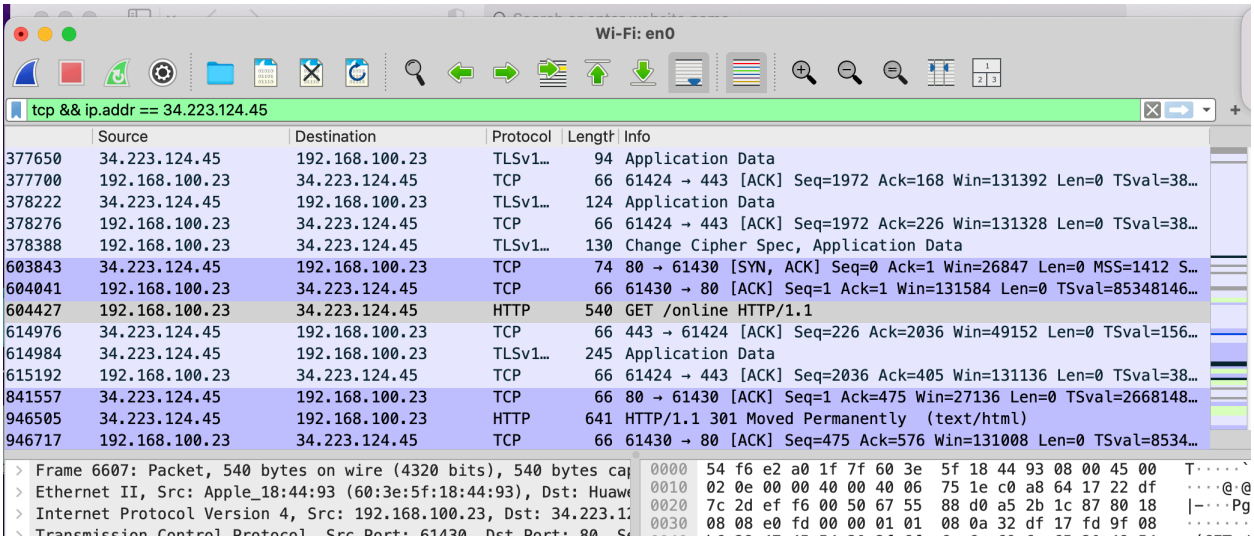


Figure 3: TCP connection establishment showing SYN packets.

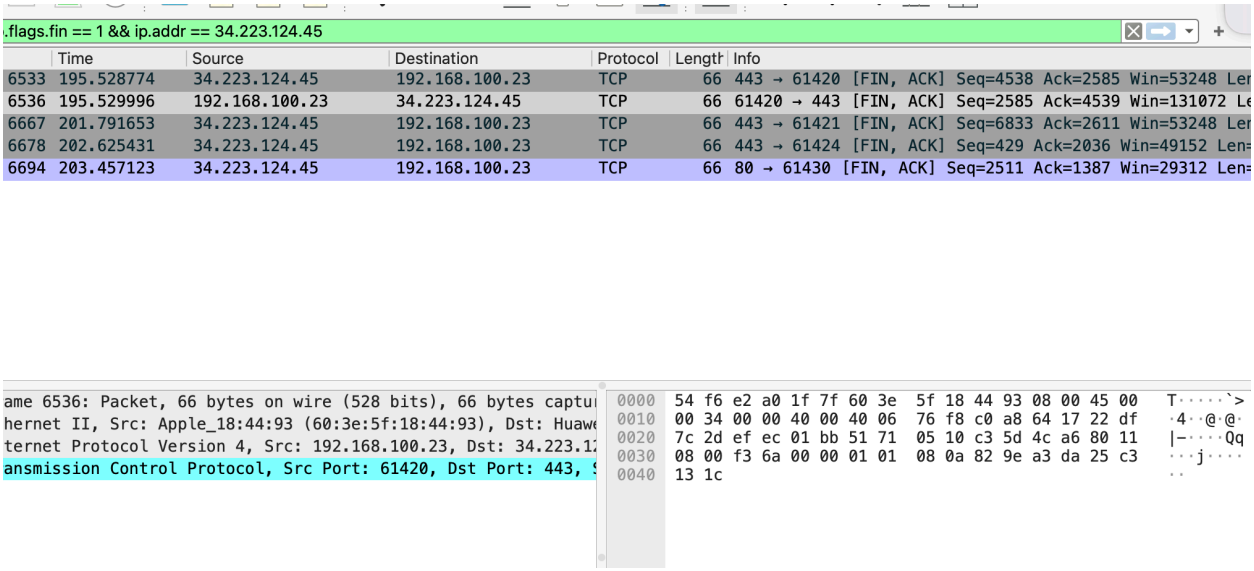


Figure 4: TCP connection termination showing FIN and ACK packets.

Part 3: UDP Analysis:

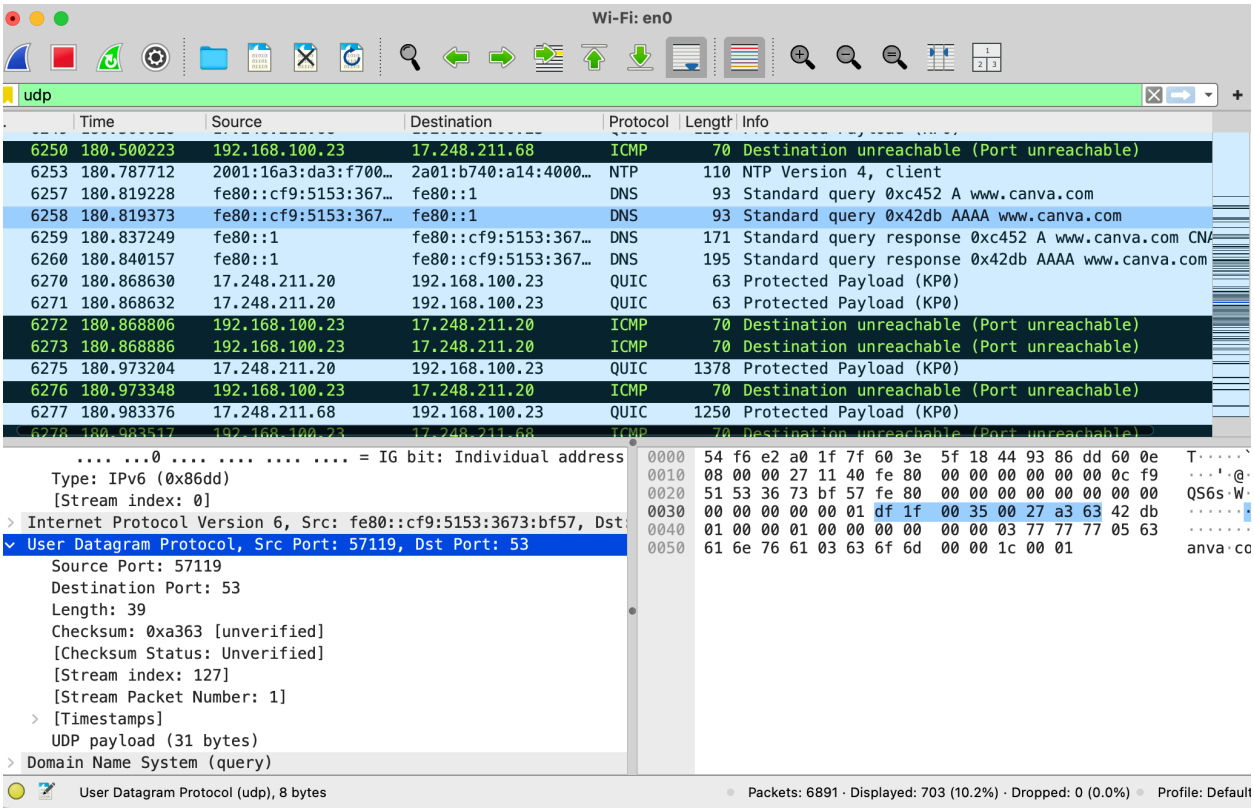


Figure 5: UDP packet details captured using Wireshark.

Part 4: Comparing TCP and UDP:

	TCP or UDP	**Reasons**
Reliability and Connection Establishment	**TCP**	TCP is connection-oriented and provides reliable data transmission using acknowledgments and retransmissions.
Data Integrity and Ordering	**TCP**	TCP ensures that data arrives correctly and in the proper order using sequence numbers and error checking.

	TCP	**UDP**
Use cases	Web browsing, email, and file transfer	Video streaming, VoIP, and online gaming.
Performance	Slower due to connection setup, acknowledgments, and error checking.	Faster with lower overhead because it does not establish a connection or wait for acknowledgments.